

Devasheesh Mishra

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SUMMARY

Founder and CTO of Recallr AI and an AI/ML and backend engineer specializing in memory for AI systems. My interests include AI agents, speech AI, and personal assistants. Authored three studies on scalable long-term memory, retrieval-free parametric memory, and hallucination dynamics in language models. I previously founded an AI home-automation startup that was later acquired. Attended Founders, Inc.'s Canopy 2026 and Y Combinator's AI Startup School in SF.

WORK EXPERIENCE

Founder and CTO

Recallr AI Inc.

January 2025 – Present

San Francisco, CA, USA — On-site

- Built Recallr, a persistent, queryable long-term memory layer for conversational AI systems that retains facts, preferences, relationships, and decisions across multiple conversations while preserving the source of each memory. Architected its ingestion pipeline, evolving knowledge graph, semantic retrieval, temporal reasoning, knowledge-update handling, and configurable merge and conflict-resolution rules.
- Built and ran an LLM evaluation and benchmarking pipeline for Recallr on [LongMemEval](#), achieving 97.5% overall accuracy, including 97.0% temporal-reasoning accuracy and 97.4% knowledge-update accuracy. Optimized inference and retrieval performance to P95 latency of 408 ms for Low-Latency recall, 1.575 seconds for Balanced recall, and 8.619 seconds for Agentic recall.
- Evolved Recallr into the intelligence and memory layer for private capital, transforming fragmented deal memos, data rooms, diligence, meeting transcripts, partner notes, filings, returns, and investment-committee history into a continuously updated, queryable decision graph that preserves how a firm's investment judgment evolves over time. Generated over \$300,000 in revenue across eight institutional pilots and raised \$200,000 in angel investment.

Canopy 2026

Founders, Inc.

March 2026 – May 2026

San Francisco, CA, USA — On-site

- Selected as one of 100 teams globally for Canopy 2026, a five-week on-site builder residency at Founders, Inc.'s San Francisco campus.
- Built and commercialized Recallr's memory layer for conversational AI systems during the program, acquiring multiple clients and generating approximately \$15,000 in revenue.

AI/ML Engineer

MIRA

September 2025 – November 2025

San Francisco, CA, USA — Remote

- Real-Time Streaming Speech-to-Text Pipeline: Designed and implemented a low-latency, WebSocket-based streaming transcription system using Soniox STT models (stt-rt-preview-v2), with real-time speaker diarization, silence detection, and multilingual translation. Built a comprehensive benchmarking framework to evaluate ASR performance across Soniox, Deepgram, Google, and AssemblyAI using word error rate, latency, and speaker-attribution accuracy, then used the results to guide model selection and system design.
- Voice Fingerprinting and Speaker Verification: Engineered an end-to-end voice-biometric pipeline covering user enrollment, audio serialization, PostgreSQL storage, and real-time speaker identification. Used SpeechBrain's ECAPA-TDNN model trained on VoxCeleb to generate and compare speaker embeddings. Deployed the system as a standalone FastAPI microservice on AWS ECS, with parallel cosine-similarity verification across diarized speaker segments, identifying the enrolled user among ambient speakers with under 200 ms inference latency.
- Long-Term Memory Extraction from Personal Data: Built an LLM-powered memory-extraction pipeline using Gemini 2.0 Flash to process a user's Gmail corpus and identify durable signals such as personality traits, expertise, preferences, and behavioral patterns. Structured the extracted information into long-term user knowledge representations, enabling AI agents to maintain context across conversations.

Founder and CEO

Stapes AI

May 2024 – January 2025

New Delhi, India — On-site

- Founded an AI home-automation company, raised \$100,000 in angel investment, shipped v1.0 within one month, and onboarded 50+ beta testers; the company was later acquired.
- Developed a home-automation platform integrating smart TVs, Fire TV devices, and switchboards, with control through voice and mobile apps; designed custom PCBs and ESP32 firmware for the device stack.
- Admitted to Buildspace's Nights & Weekends S5 program, backed by Y Combinator and a16z; subsequently white-labeled the home-automation technology for other operators.

AI/ML Engineer Intern

Proeffico Solutions Private Ltd.

August 2024 – September 2024

Noida, India — On-site

- Engineered and deployed a real-time shoplifting-detection system across retail locations, validating it with live on-site video feeds for a global client.
- Developed RDBMS Chat, an internal conversational AI tool that enabled non-technical staff to query complex relational databases in natural language, improving access to operational data and insights.
- Supported internal servers and virtual machines, helping maintain operational stability for backend systems across multiple projects.

Builder, Nights & Weekends S5

Buildspace

June 2024 – August 2024

San Francisco, CA, USA — Remote

- Joined a startup program backed by Y Combinator and a16z, focused on rapid product development for Stapes AI.
- Accelerated Stapes AI through community feedback, mentorship, rapid prototyping, and successive product iterations.
- Applied Python, Flutter, and embedded-systems development with ESP-IDF to refine the product's software and hardware stack.

Technical Lead

GeeksforGeeks

April 2024 – February 2026

New Delhi, India — On-site

- Led 10+ GeeksforGeeks machine-learning and coding workshops across SRM, teaching ML/DL fundamentals to 80+ students and reaching 200+ students through coding support.
- Managed eight technical contributors, project timelines, troubleshooting, and server operations.

Core Technical Team Member

GeeksforGeeks

October 2022 – April 2024

New Delhi, India — On-site

- Contributed as a core member while organizing the Phoenix Hackathon, where I also won first place.
- Organized Hack-Innovate, a two-day hackathon that attracted 300+ participants and showcased more than 50 project submissions.

RESEARCH

A Name Is Not a Memory: Relational Binding in Continual Parametric Memory for

August 2026 – Present

Conversational Agents (In Progress)

- **Goal:** Develop retrieval-free long-term memory for conversational agents by encoding user-specific knowledge in persistent LoRA adapters on a frozen language model; evaluate how parametric memory compares with retrieval-based systems such as Recallr AI on retention, identity binding, continual updates, and cross-user interference; and explore a scalable multi-LoRA architecture inspired by Mixture-of-Experts routing.

Hallucination Dynamics in Small Language Models:

January 2026 – Present

A Large-Scale Empirical Study (In Progress)

- **Goal:** Determine how sampling temperature affects hallucination behavior in small language models across model scales, quantization levels, and factual domains.
- **Approach:** Collected over 10 million data samples from SLMs including Qwen2.5-0.5B/1.5B/3B/7B, Qwen3-1.7B/4B/8B, Llama-3.2-1B/3B, Gemma-3-4B, and Mistral-7B. Tested $T \in [0, 1]$ at $\Delta T = 0.1$ on HaluEval-QA and True/False across politics, geography, history, science, health, cities, companies, facts, and other domains.
- **Result:** Identified a non-monotonic scale-temperature relationship: sub-1.5B SLMs often collapsed into one-class prediction priors at low temperature, so added sampling entropy could reduce hallucination error by weakening systematic bias; near the 1.5B competence boundary, higher temperature instead amplified factual instability, especially under 4-bit quantization, while many 3B–8B models remained comparatively stable. Effects varied by topic and were concentrated on marginal knowledge, showing that hallucination risk depends on the interaction of scale, calibration, quantization, and domain rather than temperature alone.

Recallr AI: Scalable Architecture for Dynamic Long-Term Memory in Conversational

April 2025 – Present

AI Agents (In Progress)

- **Goal:** Build scalable long-term memory for conversational AI that preserves evolving user state across sessions without adding latency to real-time responses.
- **Approach:** Built a versioned knowledge graph through decoupled asynchronous curation and synchronous retrieval. Its decision logic classifies new evidence as redundant, novel, additive, temporally superseding, expired, or directly conflicting; dual timestamps separate event time from discussion time, immutable version chains preserve state evolution, and a Git-inspired human-in-the-loop protocol resolves contradictions. Auto-Recall routes queries across Low-Latency, Balanced, and Agentic retrieval with adaptive graph traversal.
- **Result:** Benchmarked Recallr on the LongMemEval Oracle tier. Agentic Recall achieved 98.9% accuracy—32.7 percentage points above the nearest competitor—with 100% on knowledge updates, 99.2% on temporal reasoning, and 98.5% on multi-session questions; Low-Latency Recall retained 92.6% accuracy at 396 ms P95, approximately $4.5\times$ faster at P95 than the fastest competing system.

PROJECTS

Codex Proxy | Python, FastAPI, PostgreSQL, Docker

- Built a self-hosted, OpenAI-compatible gateway that pools authorized Codex subscription accounts behind one endpoint. Implemented quota-aware priority routing, asynchronous concurrency control, fault-tolerant failover, per-user model and budget policies, encrypted credentials, observability, usage accounting, request archives, notifications, and an operations dashboard. Processed more than 20 billion tokens across approximately 70,000 requests in three months.

Claude Code Proxy | Python, FastAPI, PostgreSQL, Docker

- Built a self-hosted, Anthropic-compatible gateway that pools authorized Claude subscription accounts behind one endpoint. Implemented quota-aware priority routing, asynchronous concurrency control, fault-tolerant failover, per-user model and budget policies, encrypted credentials, observability, usage accounting, request archives, notifications, and an operations dashboard.

Universal TV Remote for Mac | Swift, SwiftUI, Bonjour, SSDP, WebSockets

- Built a native, approximately 1 MB SwiftUI menu-bar remote that automatically discovers and controls Fire TV, Android and Google TV, Samsung, LG, and Roku devices without configuration or cloud services. Unified brand-specific REST, TLS and protobuf, WebSocket, and ECP protocols behind a common driver interface, with one-time pairing, persistent tokens, Wake-on-LAN, keyboard input, and app shortcuts.

yc-api | Python, Playwright, OpenAI, TypeScript, GitHub Actions

- Built a free, authentication-free JSON API and typed Python and TypeScript SDKs covering more than 6,000 Y Combinator companies across 50 batches. A GitHub Actions CI/CD pipeline automatically refreshes and republishes 50+ fields every day, including founders, verified emails, jobs, news, and launches, and processes 255 public application and Demo Day videos into timestamped, speaker-diarized transcripts.

ACHIEVEMENTS

Y Combinator’s AI Startup School: Chosen as one of 2,000 CS students worldwide to attend YC’s first AI Startup School in San Francisco.	(2025)
Google InnoSprint Hackathon: First runner-up.	(2023)
Phoenix Hackathon: Won first place.	(2022)
Code-A-Thon: First runner-up.	(2022)
Live-Project Competition: Second runner-up.	(2023)
Smart India Hackathon: First runner-up at the college level.	(2023)

SKILLS

Programming: Python, C/C++, JavaScript/TypeScript
AI/ML: PyTorch, Transformers, LLMs, Agentic AI, RAG, LLM Evaluation, Fine-tuning/LoRA, LiteLLM, Pandas, NumPy, Matplotlib
ML Systems: Model Serving & Inference, MLOps, Low-Latency Systems, GPU Inference Optimization, Data Pipelines
Backend & Data: Distributed Systems, System Design, FastAPI, Pydantic, REST APIs, WebSockets, Async/Concurrency, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ, Vector Databases (Milvus), Knowledge Graphs (Neo4j), MinIO
Cloud & Infrastructure: AWS (ECS, ECR, EFS), Docker, Docker Compose, Kubernetes (K3s), Linux, Portainer, Traefik, Harbor, Terraform
CI/CD & Reliability: Git, GitHub, GitLab CI, GitHub Actions, Pytest, Selenium, High Availability, Fault Tolerance, Prometheus, Grafana, Loki, Promtail, Alertmanager, Elastic Stack (Elasticsearch, Kibana, Filebeat)
Embedded Systems: ESP-IDF, PlatformIO, Arduino, ESP32/ESP8266, ESP32-CAM, NodeMCU, Raspberry Pi

INTEREST

Homelab Overlord	September 2020 – Present
- Designed and maintain a self-hosted cloud and AI compute environment on an Inspur server. It combines bare-metal virtualization, distributed GPU compute, Kubernetes orchestration, private networking, storage, CI/CD, and observability, providing a production-like platform for personal services and ML workloads. - The server is equipped with a 64-core CPU, 128 GB RAM, redundant network interfaces, four NVIDIA Tesla P40 GPUs with 24 GB VRAM each for local LLM inference, and two NVIDIA RTX 3060 GPUs for image generation, ASR, and TTS workloads. Storage includes five 2 TB HDDs, with two assigned to parity and three in the Unraid array, plus two 1 TB NVMe drives, including one dedicated to the storage-pool cache. - Developed custom inference kernels and runtime optimizations for Tesla P40 hardware, improving compatibility and efficiency when running language models on older GPU architectures. - Use Proxmox VE as the bare-metal hypervisor and host Unraid in a dedicated VM as the storage server for persistent storage, backups, and large ML and project datasets. Operate a heterogeneous K3s cluster with a Raspberry Pi control-plane node, two CPU worker nodes, and two GPU-accelerated workers for ML experiments and containerized services. - Self-host GitLab for source control and CI/CD, Harbor for private Docker and OCI image storage, and Traefik for ingress routing and automated TLS termination. Use Prometheus, Grafana, and Loki for infrastructure metrics, dashboards, and centralized logging. - Run privacy-focused applications including Bitwarden, Nextcloud, Jellyfin, and Immich, with persistent storage and controlled external access. - Built and operate a local-first, Jarvis-style assistant that originated from Stapes AI and controls nearly every device and service in my home. Core processing and automation run on local hardware. - Built a custom application to monitor and control the Inspur server, available at inspur-mgmnt-app.	

CERTIFICATIONS

Data Mining	IIT, Kharagpur
Neural Networks and Deep Learning	DeepLearning.AI
PyTorch for Deep Learning Bootcamp	Udemy

EXTRACURRICULAR

Course Instructor GeeksforGeeks	September 2024 – April 2026
Assistant Course Instructor GeeksforGeeks	July 2024 – September 2024

EDUCATION

SRM Institute of Science and Technology Bachelor of Technology in Computer Science, Specialization in AI and ML; CGPA: 8.5	May 2022 – July 2026 Chennai, India
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